

CaseStudy 2: Sales Demand Forecasting System

AI Case Study – Dyashin Technosoft Pvt Ltd

Company Overview

Dyashin Technosoft Pvt Ltd is an IT Services organization providing software development, IT consulting, training, and digital solutions to clients across multiple regions.

Domain

Sales & Retail Supply Chain

Project Duration

1 Month

Business Scenario

Dyashin Technosoft Pvt Ltd has developed a retail management system that helps businesses track product sales, customer orders, and inventory.

However, retail businesses often face challenges such as:

- Overstocking of slow-moving products leading to losses
- Stockouts of high-demand products affecting customer satisfaction
- Inefficient inventory planning due to lack of demand insights

To address these challenges, the company aims to build a **Sales Demand Forecasting System** that predicts future product demand using historical sales data.

This system will help businesses make data-driven decisions and maintain optimal inventory levels.

Business Objectives

- Forecast future sales demand across different products and regions
 - Reduce overstocking and minimize inventory holding costs
 - Prevent stockouts of high-demand products
 - Improve inventory planning and supply chain efficiency
 - Identify sales trends and seasonal patterns
 - Enable better business decision-making using data-driven insights
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Project Scope

- Data cleaning and preprocessing
 - Time-series feature engineering (day, month, seasonality)
 - Training and optimizing regression-based forecasting models
 - Evaluating predicted demand against actual sales
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Models Developed

- Sales Trend Analysis Model
 - Sales Demand Predictor
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Expected Deliverables

- Python Notebook implementing forecasting model
 - Null values will be handled using imputation techniques (mean/mode/median) instead of dropping.
 - Duplicate records will be analysed and handled appropriately, not blindly removed.
 - Inconsistent categorical values will be standardized.
 - Incorrect data types will be corrected.
 - Outliers will be treated using statistical methods.
 - Visualizations of demand trends.
 - Model evaluation report including key performance metrics such as Accuracy, Precision, Recall, and F1 Score to assess prediction effectiveness.
 - Predicted sales demand for upcoming month.
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Expected Outcome

This project enhances business operations by enabling retailers to maintain optimal stock levels, reducing waste and improving product availability.